

Paranasal Sinuses

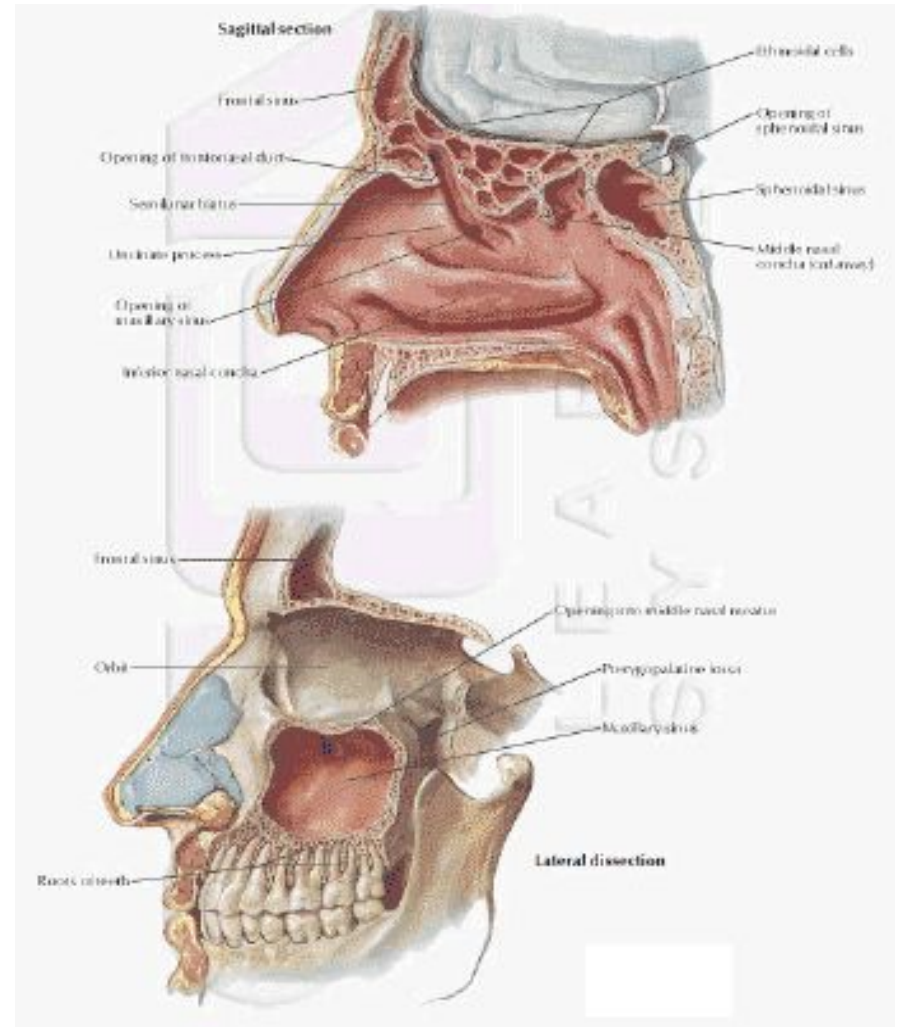
Learning objectives

At the end of the lecture the student should be able to know:

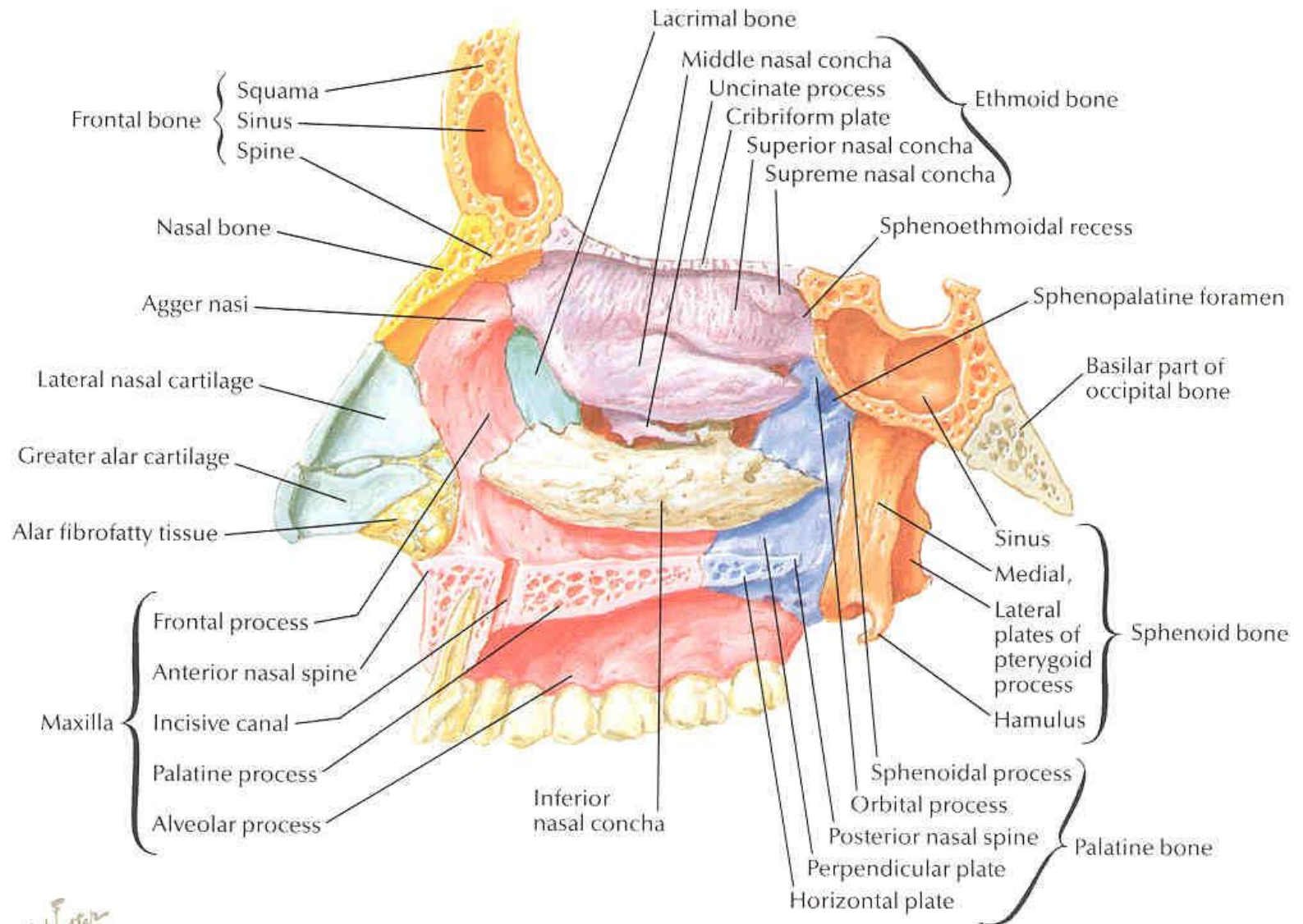
- Definition of paranasal sinuses
- Names of paranasal sinuses
- Functions of paranasal sinuses.
- Radiographic Protocols for sinuses
- Diseases of sinuses.

Paranasal sinuses

- **Paranasal sinuses** are air-filled spaces, communicating with the nasal cavity, within the bones of the skull and face.
- Divided into subgroups that are named according to the bones within which the sinuses lie



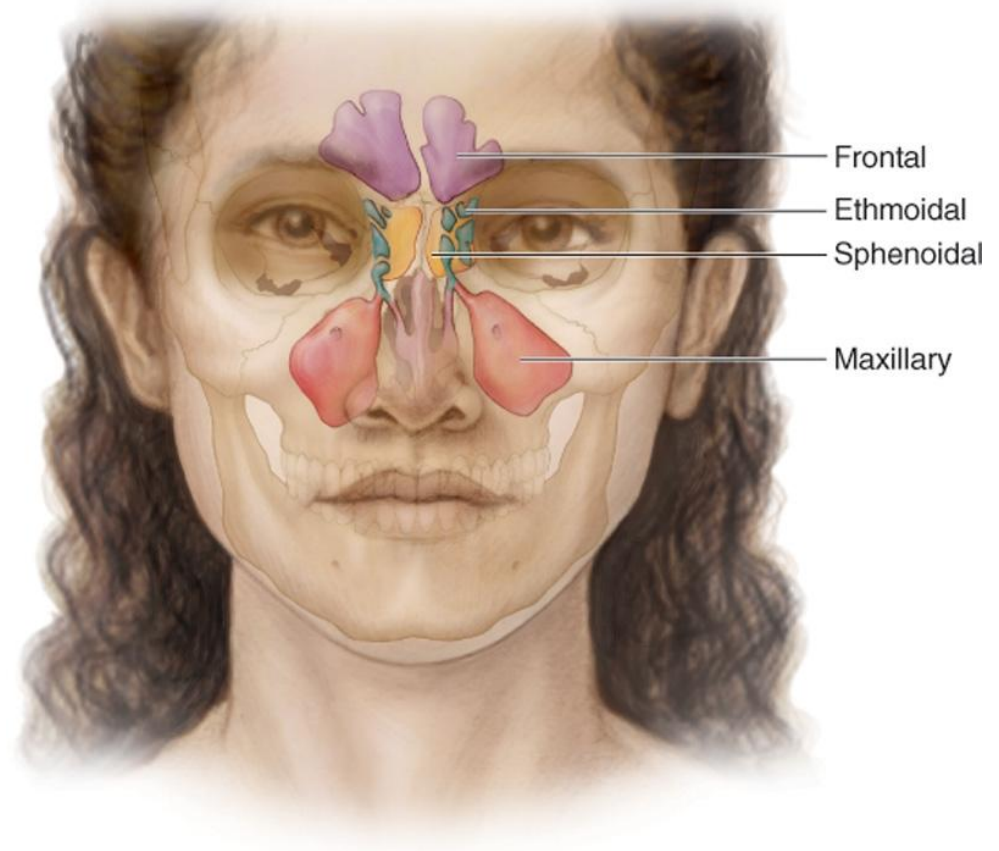
Bony structure of nose



Names of sinuses

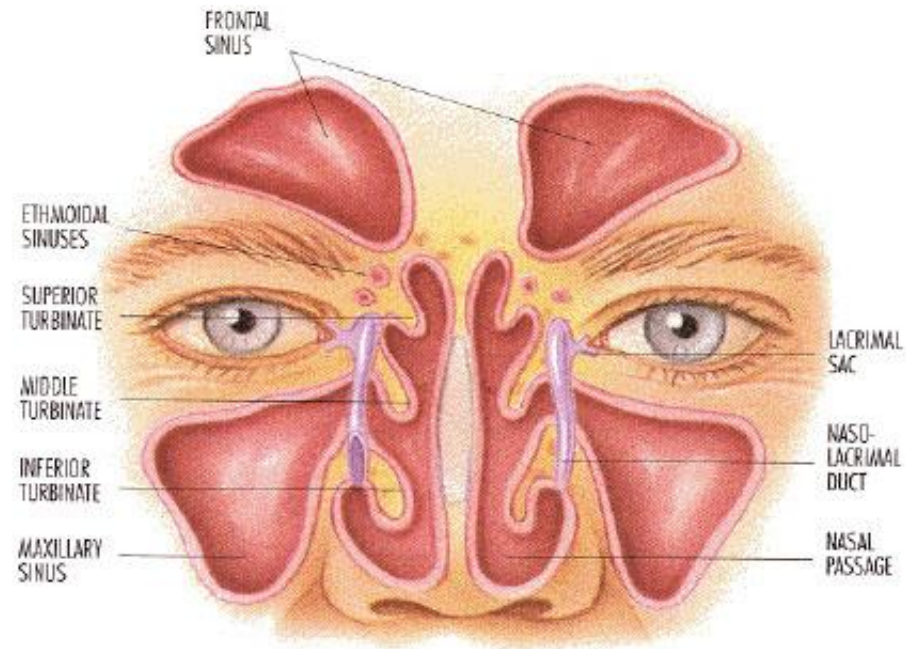
- Maxillary
- Frontal
- Ethmoid
- Sphenoid

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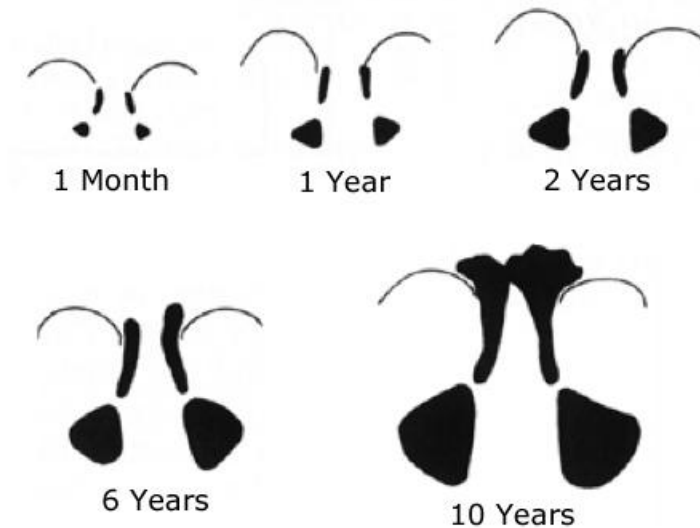
- Present at birth
- Largest sinus
- Found in body of maxilla
- Has 3 recesses:
 - Apex of maxillary sinus extends into zygomatic bone.
 - Base forms the inferior part of lateral wall of nasal cavity
 - Roof formed by the floor of the orbit.
 - Floor formed by the alveolar part of maxilla.
 - Roots of maxillary teeth (2 molars) produce conical elevations in the floor of sinus

Maxillary sinus

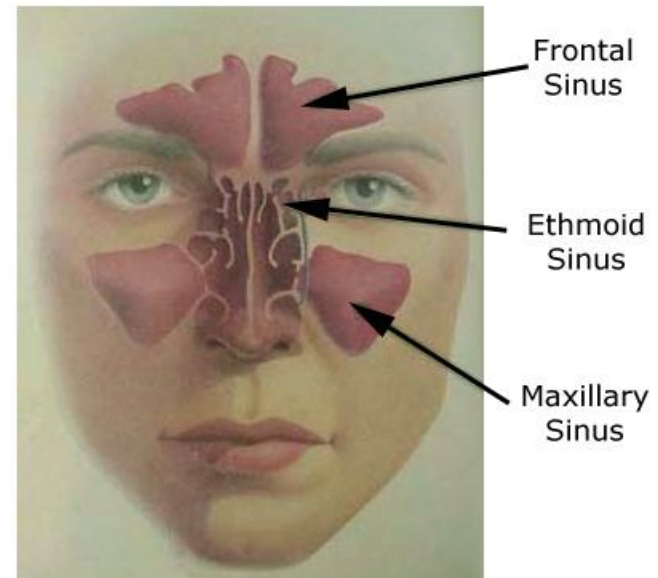


Development of sinuses

- Aerated at birth
 - **Maxillary sinuses**
- Age 6-7
 - **Frontal /sphenoidal sinuses**
- Puberty- approx 17-18 yrs
 - **Ethmoid**



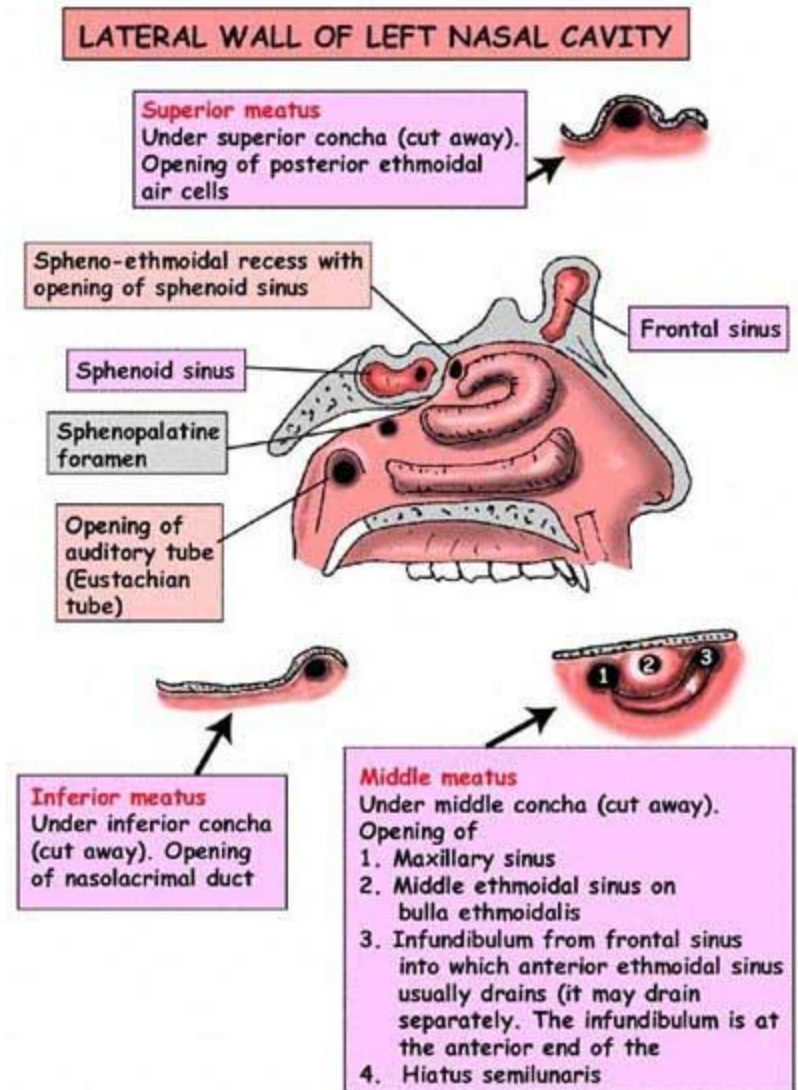
Pediatric Sinus Anatomy



Adult Sinus Anatomy
(Sphenoid Sinus not Shown)

Ostium of maxillary sinus

- The sinus communicates through an opening into the semilunar hiatus on the lateral nasal wall.
- The sinus is lined with mucoperiosteum, with cilia that beat toward the ostia

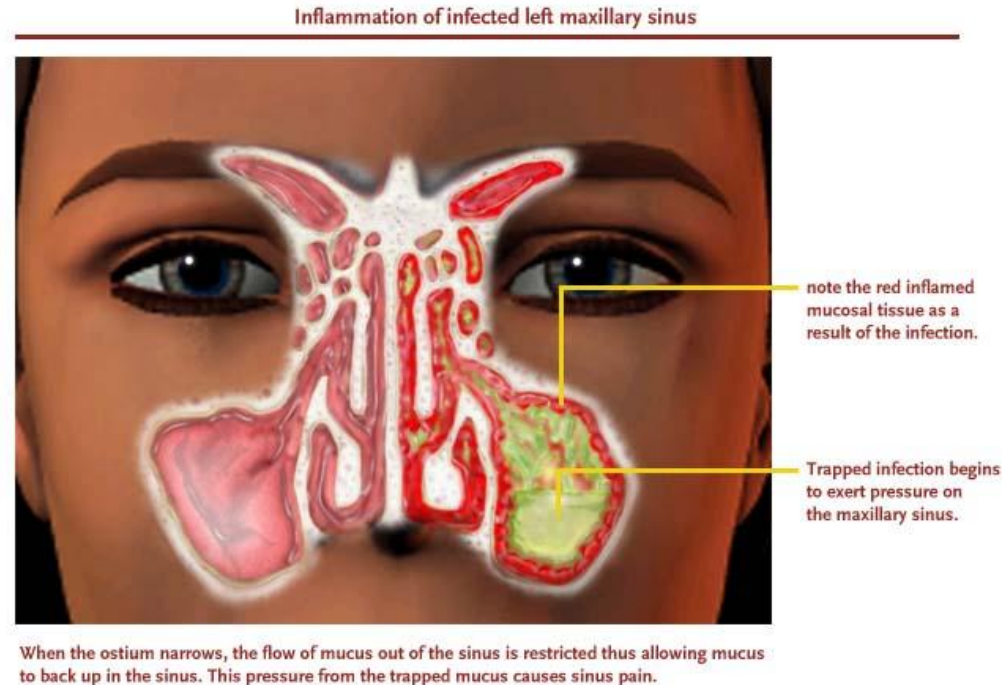


Maxillary sinusitis

- Inflammation of maxillary sinus
- Maxillary sinusitis is common due to the close anatomical relation of the frontal sinus, anterior ethmoidal air sinus and the maxillary teeth, allowing for easy spread of infection.
- Drainage orifice lies near the roof of the sinus, and so the maxillary sinus does not drain at all well, and infection develops more easily.

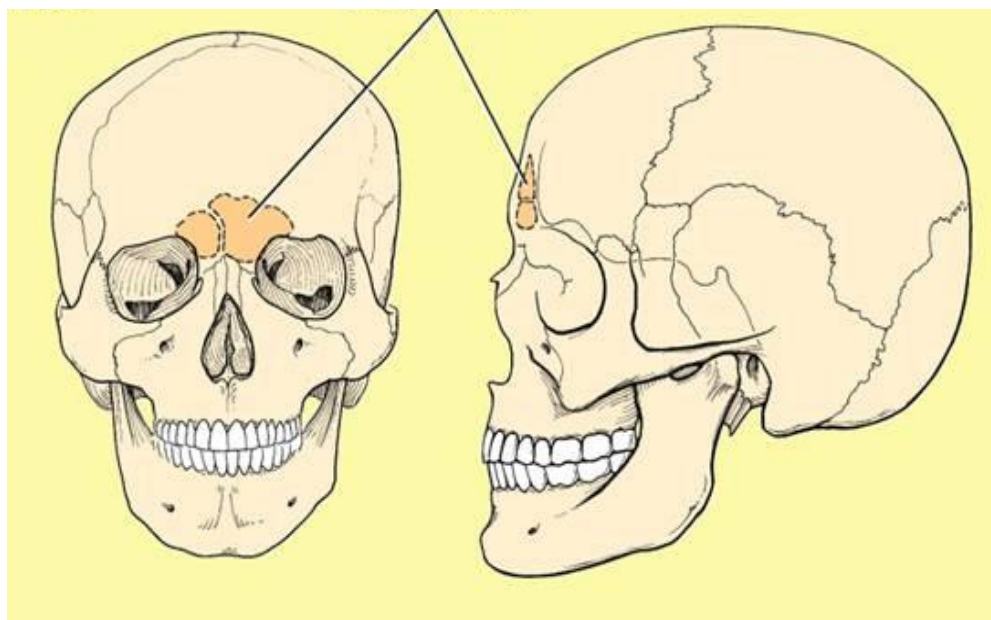
Treatment

- Broad spectrum antibiotic



Frontal sinus

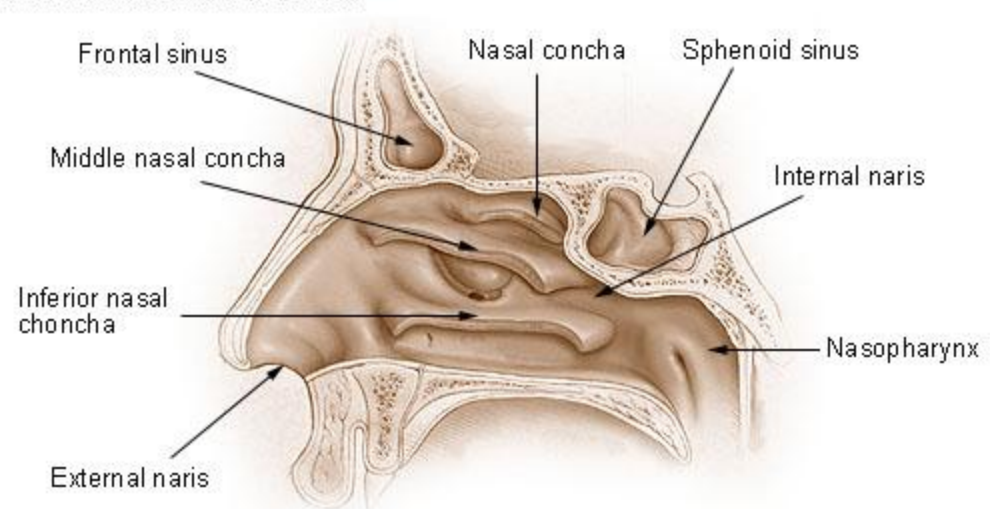
- Second largest sinuses
- Number varies
- Situated behind the superciliary arches,
- Absent at birth, reaching their full size after puberty.
- Frontal sinuses are rarely symmetrical and the septum between them frequently deviates to one or other side of the middle line



Opening of frontal sinus

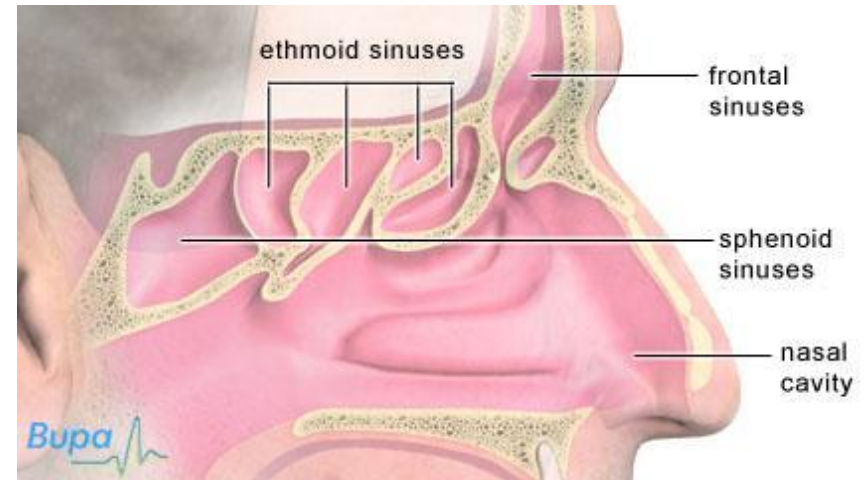
- Opens into the anterior part of the corresponding middle meatus of the nose through the frontonasal duct.
- Each frontal sinus opens into the anterior part of the middle meatus of the nose through the frontonasal duct.
- Which opens into the hiatus semilunaris
- The **hiatus semilunaris** is a halfmoon-shaped groove in the lateral wall of the nasal cavity.

Nose and Nasal Cavities



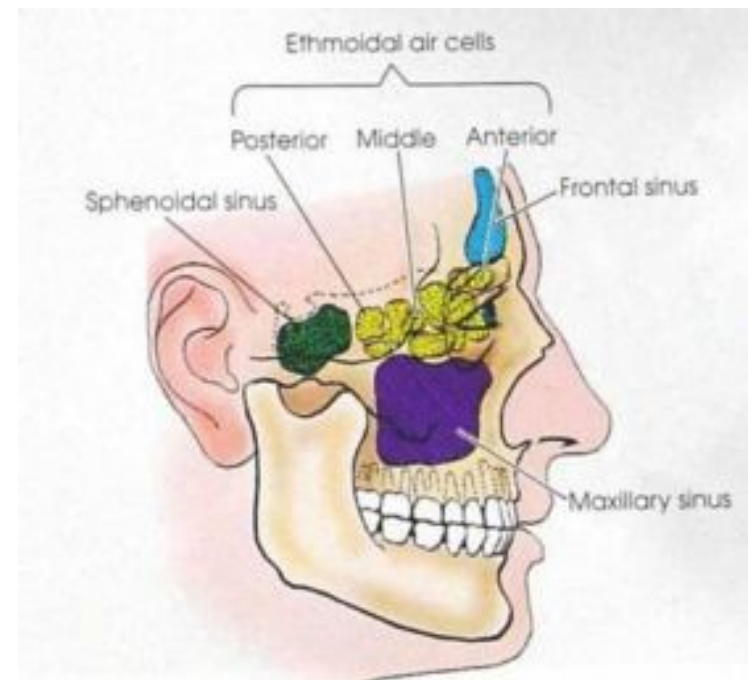
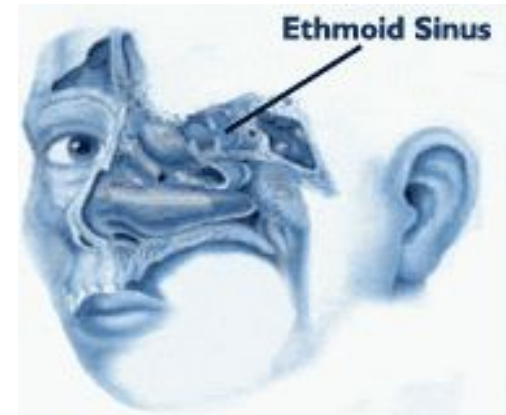
Ethmoid sinuses

- Collective name for the **ethmoidal air cells**.
- Each ethmoid sinus is an air-space enclosed within the ethmoid bone. The ethmoidal air cells consist of numerous thin-walled cavities situated in the ethmoidal labyrinth and completed by the frontal, maxilla, lacrimal, sphenoidal, and palatine bones.
- They lie between the orbits, above the nasal cavity and are separated from these cavities by thin bony laminae.



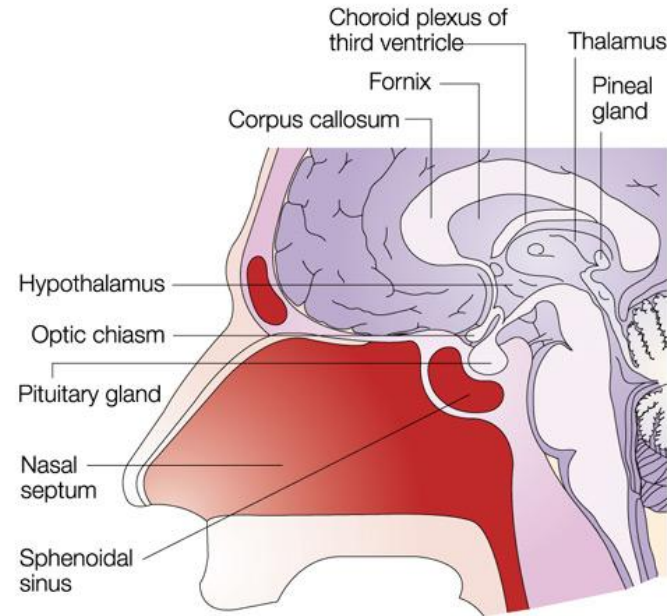
Ethmoid sinuses opening

- Anterior, middle, and posterior ethmoid sinuses.
- **Posterior** drains into the superior meatus under cover of the superior nasal concha; sometimes one or more opens into the sphenoidal sinus.
- **Middle** drains into the middle meatus of the nose on or above the bulla ethmoidalis.
- **Anterior** drains into the middle meatus of the nose by way of the infundibulum.



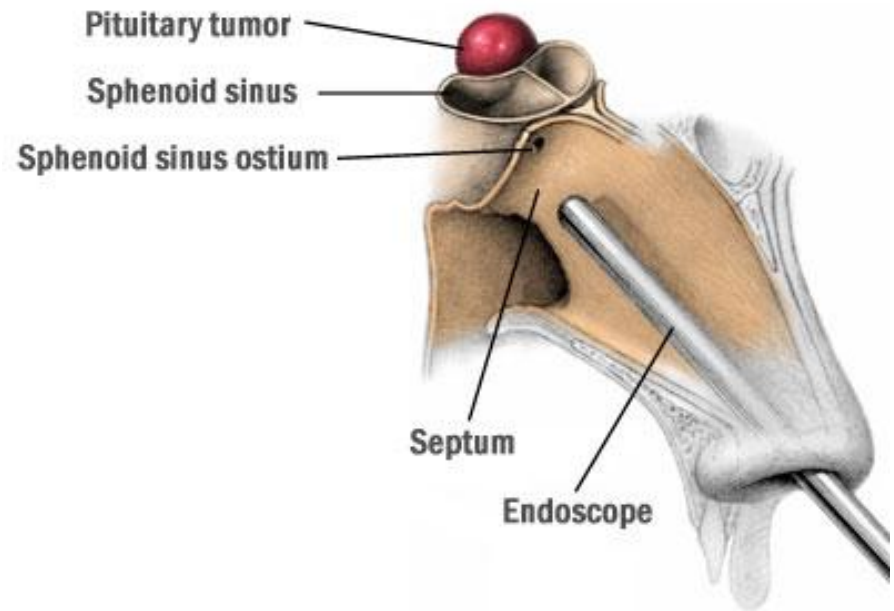
Sphenoid sinus

- They are present as minute cavities at birth, but their main development takes place after puberty.
- Present within the body of sphenoid.
- They are rarely symmetrical due to the lateral displacement of the intervening septum.
- When exceptionally large they may extend into the basilar part of the occipital bone.
- Pituitary gland can be approached through nose bypassing the sphenoid sinus.



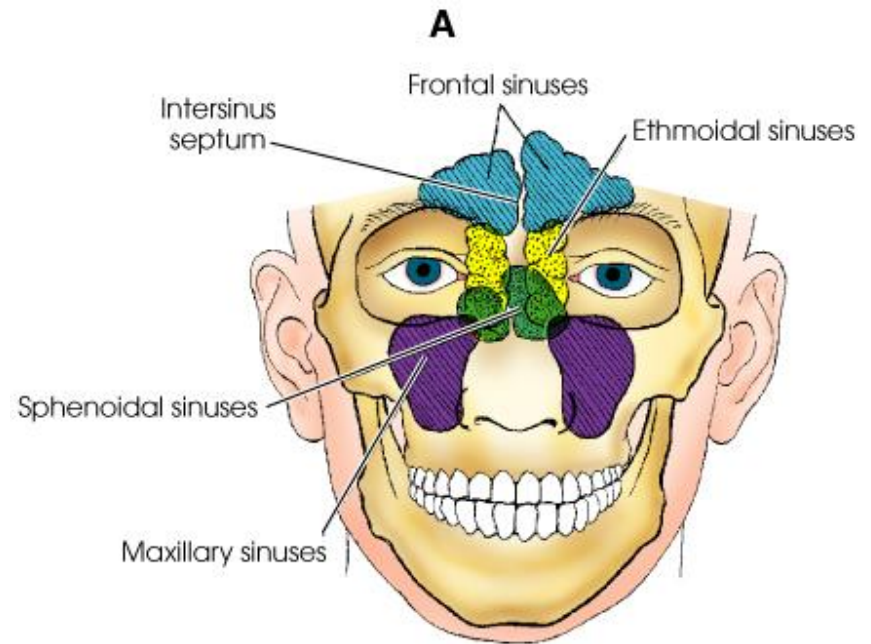
Opening of sphenoid sinus

- Each sinus opens into the roof of the nasal cavity via apertures on the posterior wall of the sphenoethmoidal recess.
- A potential complication of sphenoid sinusitis is cavernous sinus thrombosis.



Function of paranasal sinuses

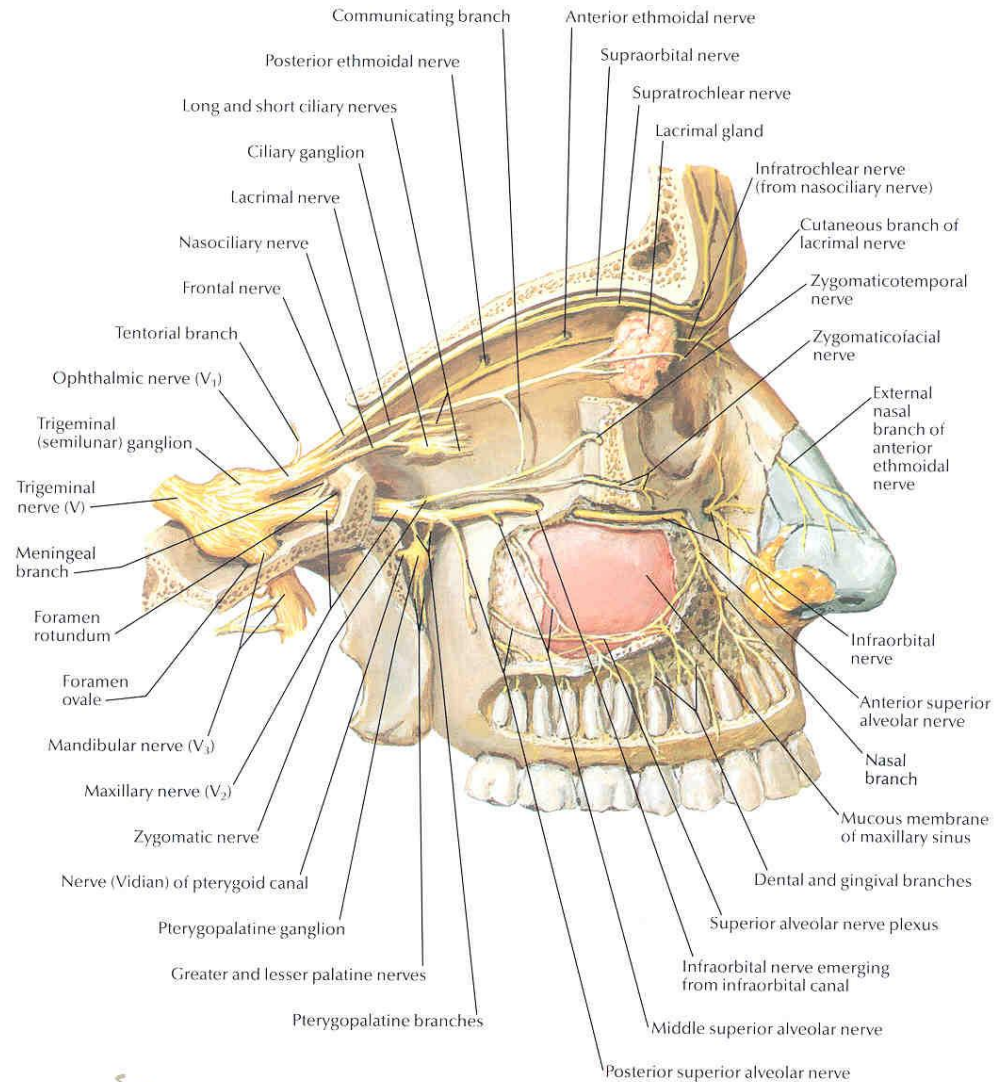
- Resonating chamber for voice
- Decrease weight of skull
- Warm & moisten air
- Shock absorbers
- Immune system



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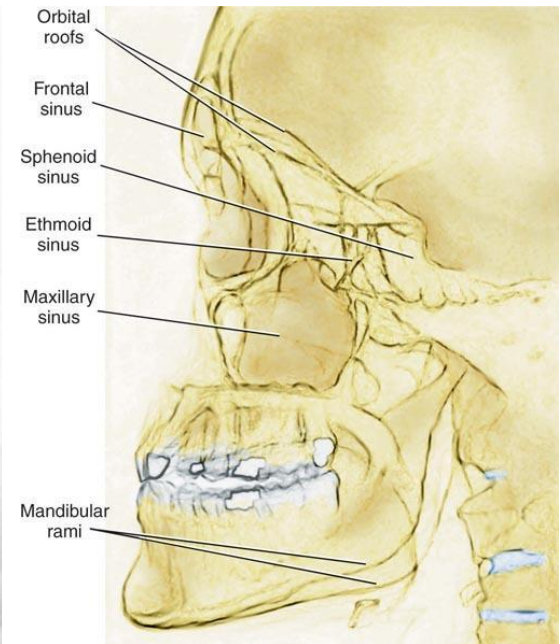
Innervation

Maxillary - Superior
alveolar nerve branch
of maxillary nerve.

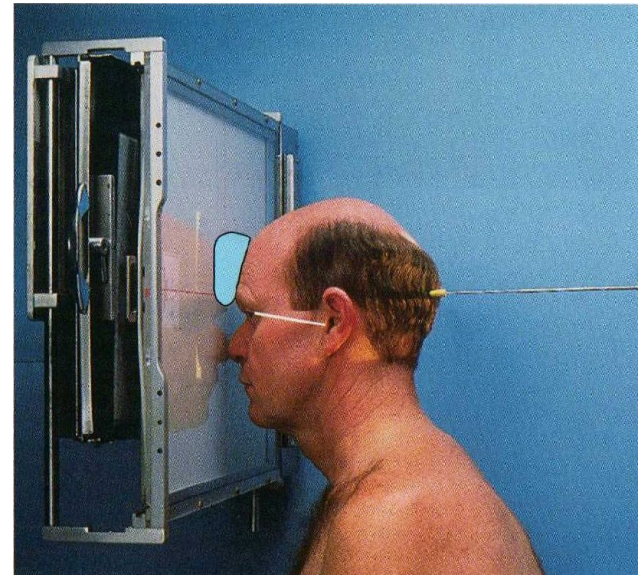
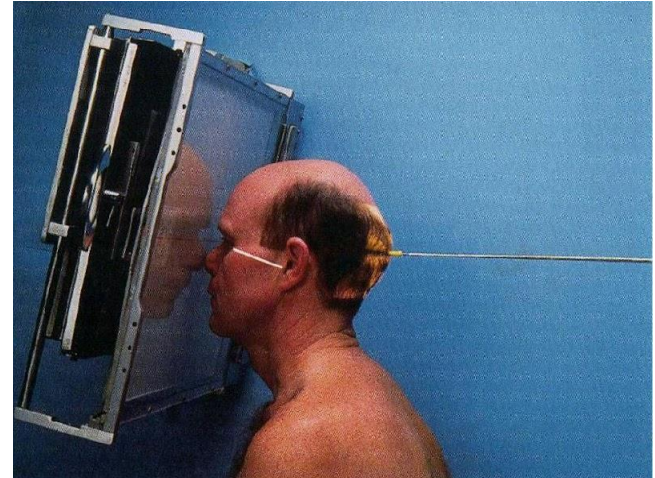


Radiology

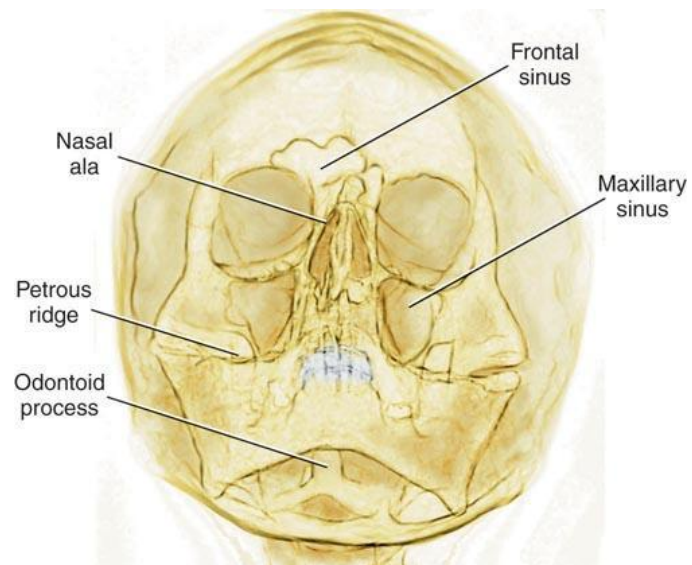
- Lateral Radiograph
- All 4 sinuses
 - Sphenoid of primary interest



- (PA) caldwell.
- Angled and vertical.
- Frontal & ethmoid air cells



- Waters view.
- For maxillary sinus



- Open mouth waters
- Maxillary and sphenoid.

